CAMS Nerd Herd InvenTeam

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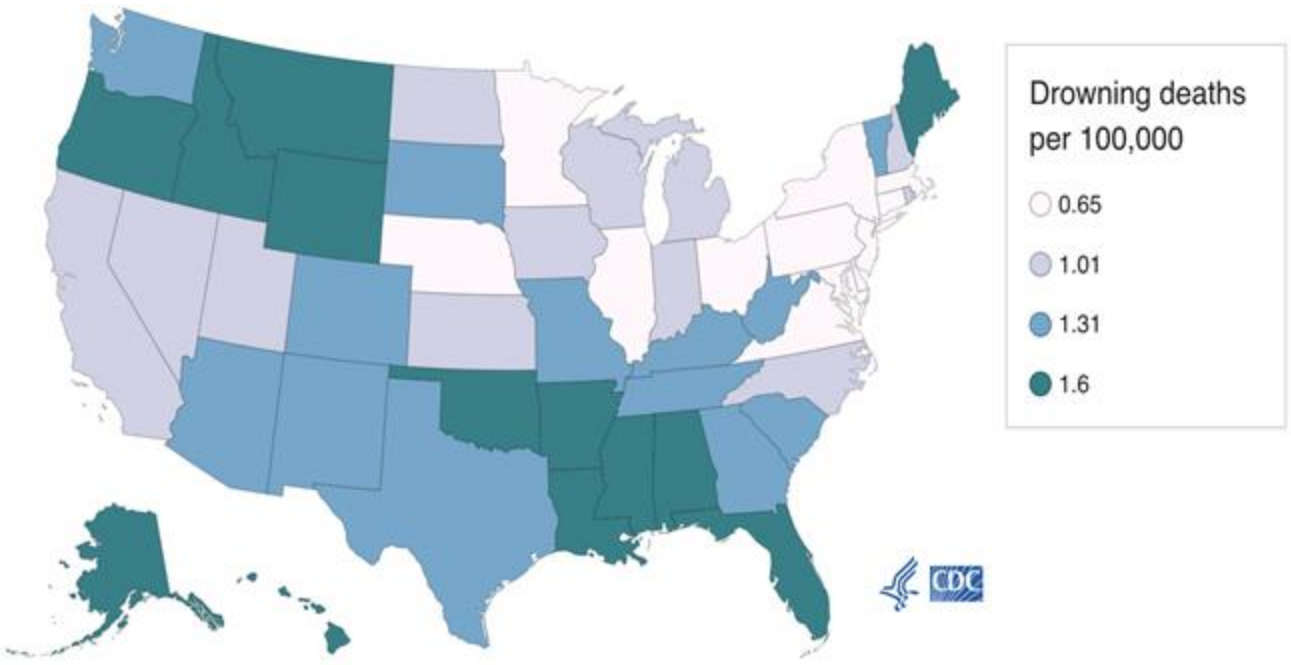
InvenTeams
California Academy
of Mathematics and Science
Carson, California
The Lemelson Foundation
MIT
School of Engineering

Our Purpose



USA Drowning deaths in the year 2019

- 2nd leading cause of unintentional death among children (CDC)
- 236,000 deaths annually in the world (WHO)
- High density of local ports and waters in California

Solution Design

01

Usable by children 4-14

02

Communicate wirelessly to a device

03

Sense drowning with vitals

04

Auto-inflate to support intervention

Prior Art

Wristwatches cannot consistently communicate in water

Wristwatches do not use vitals to detect drowning

Flotation devices require manual activation

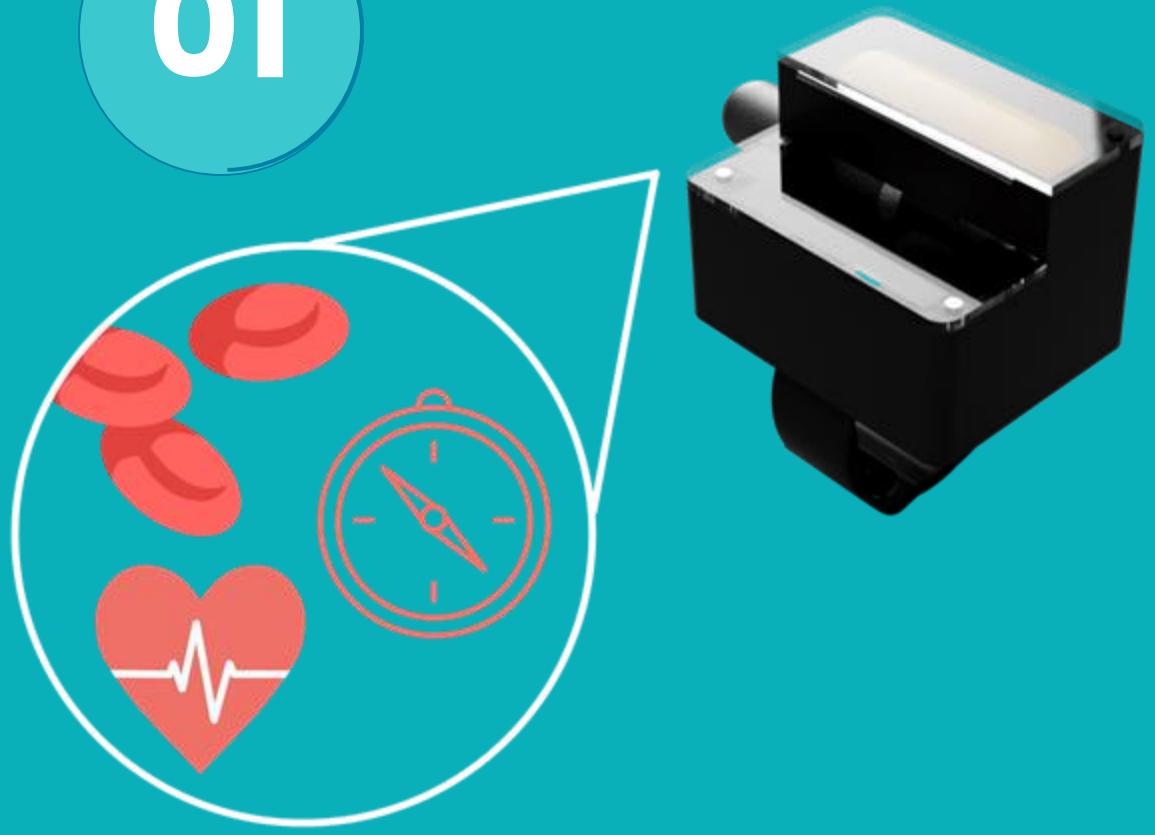
Our Invention

01 **Detect**

02 **Alert**

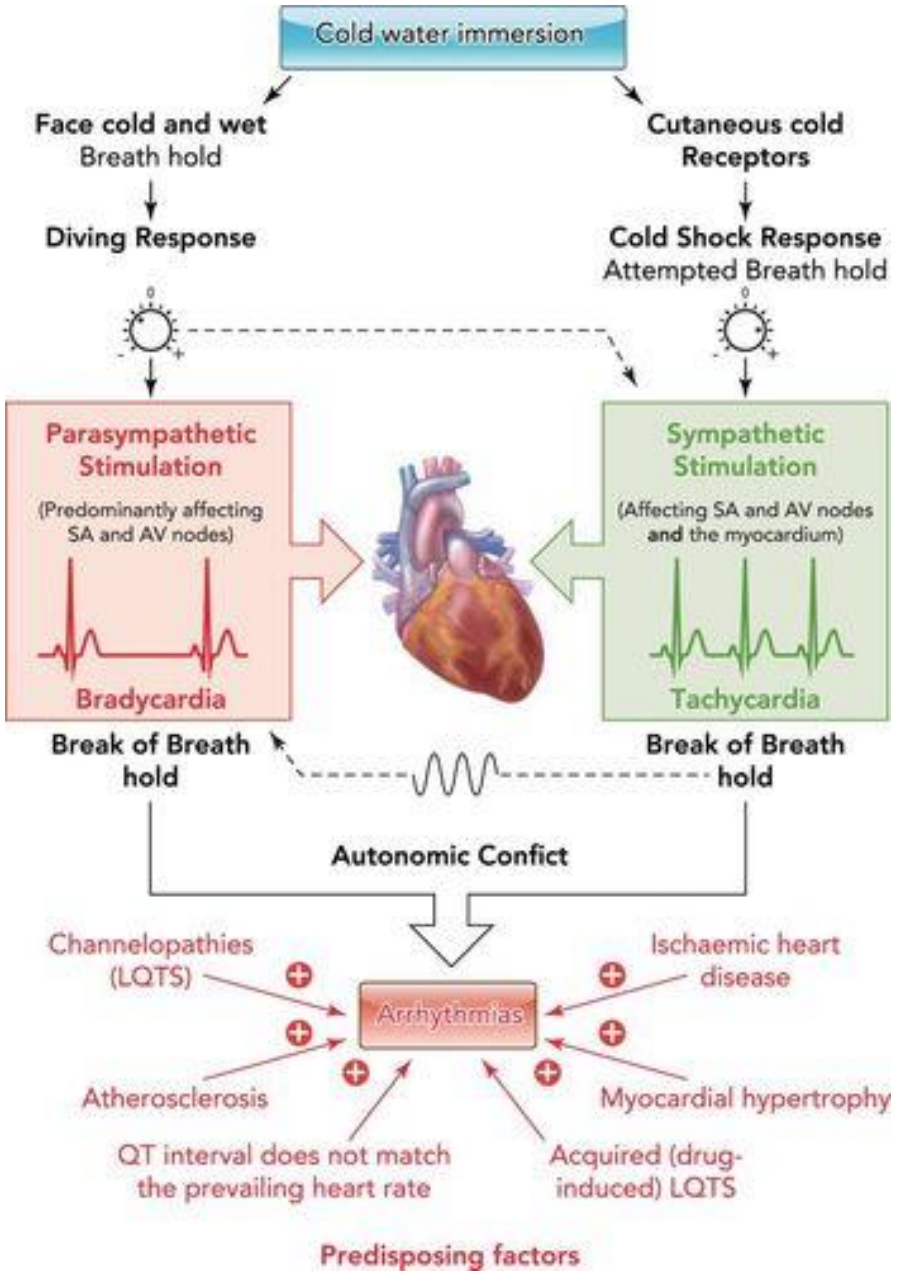
03 **Support**

01



Detect

changes in heart rate, body temperature, and blood oxygen saturation



- Heart rate detected using photoplethysmography (PPG)
- Body temperature with temperature sensor
- Blood oxygen saturation using pulse oximetry

02

Alert

parents and supervisors
of child’s status

Underwater Communication Testing

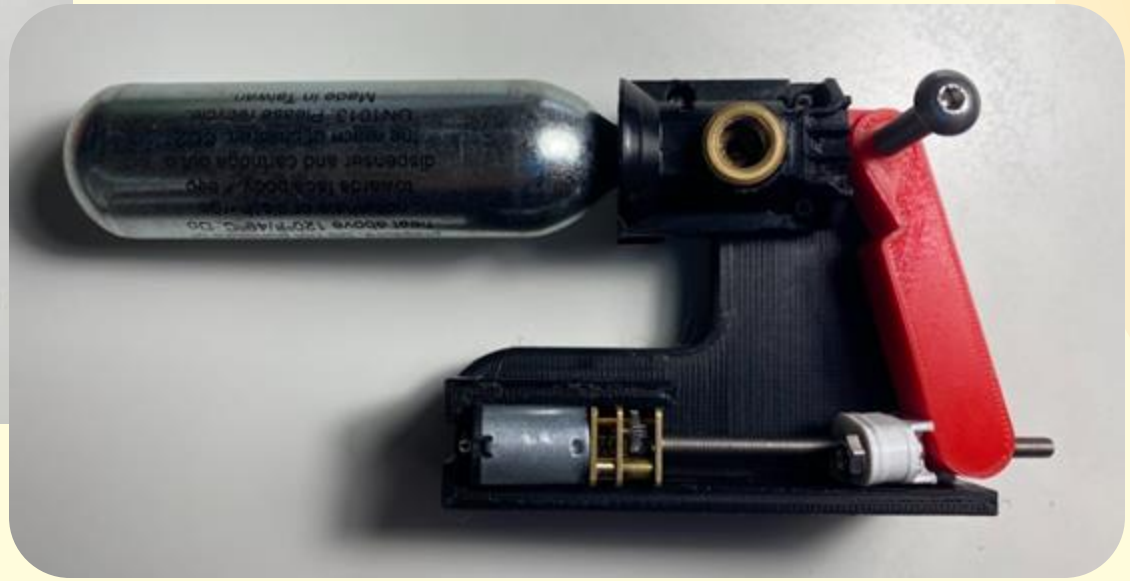
Problem:
All wireless signals are attenuated

Solution:
Wi-Fi board communication

03

Support

by deploying a flotation device



Detect Drowning

Notify Adult

Activate Motor

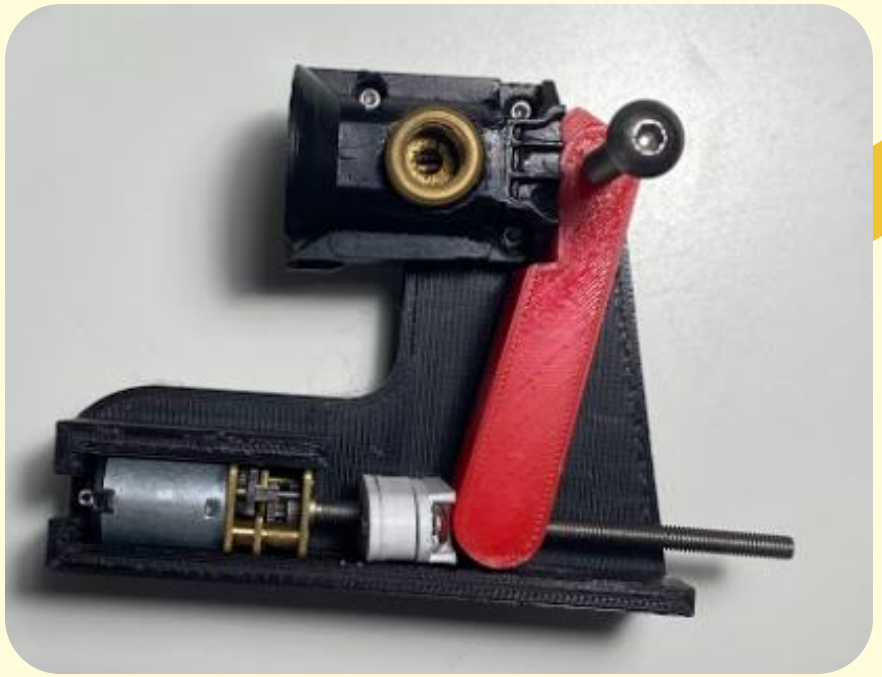
Flotation Deploys

- Small brushed motors push lever
- Replaceable air canisters
- User can hold onto the inflatable as help arrives

Invention Iteration

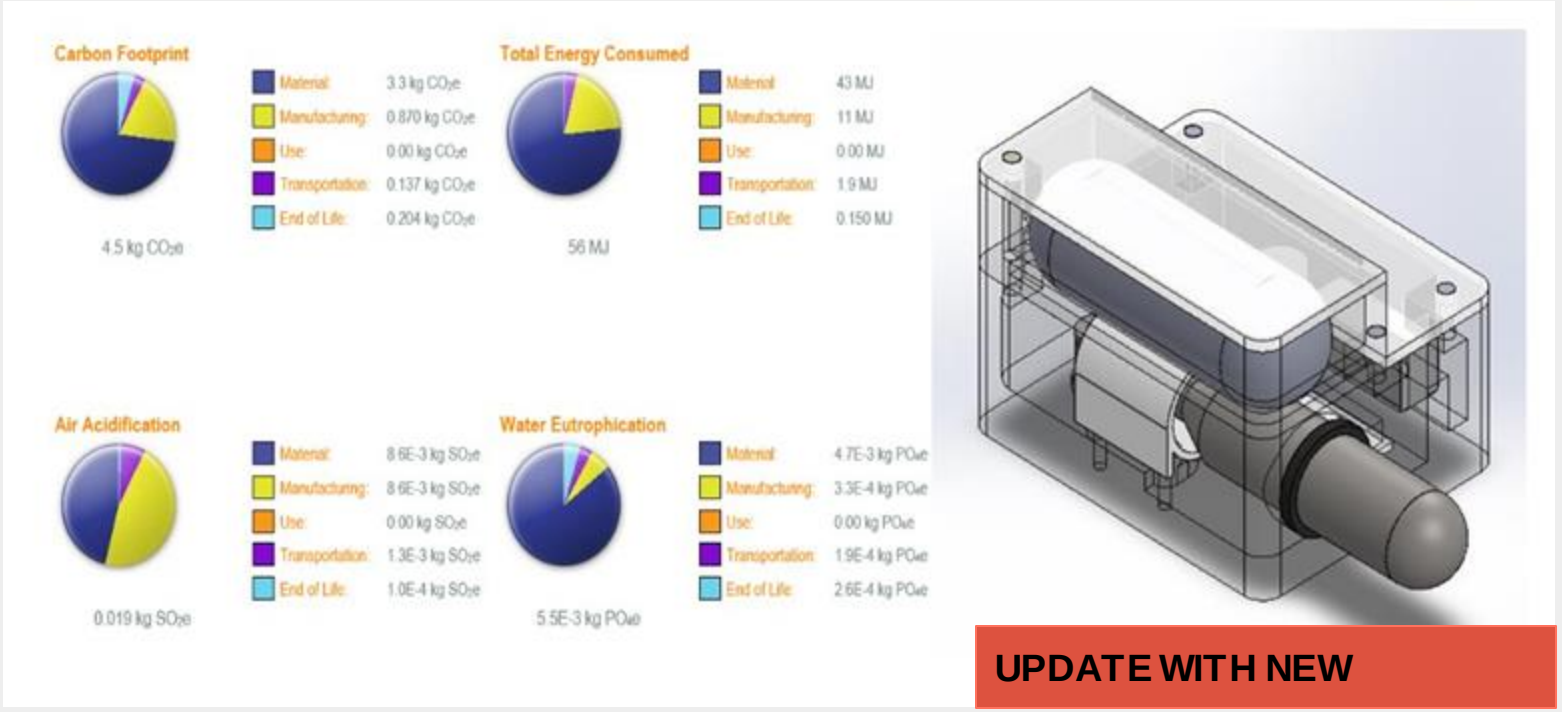
Prototype #1

Prototype #2



Prototype #3

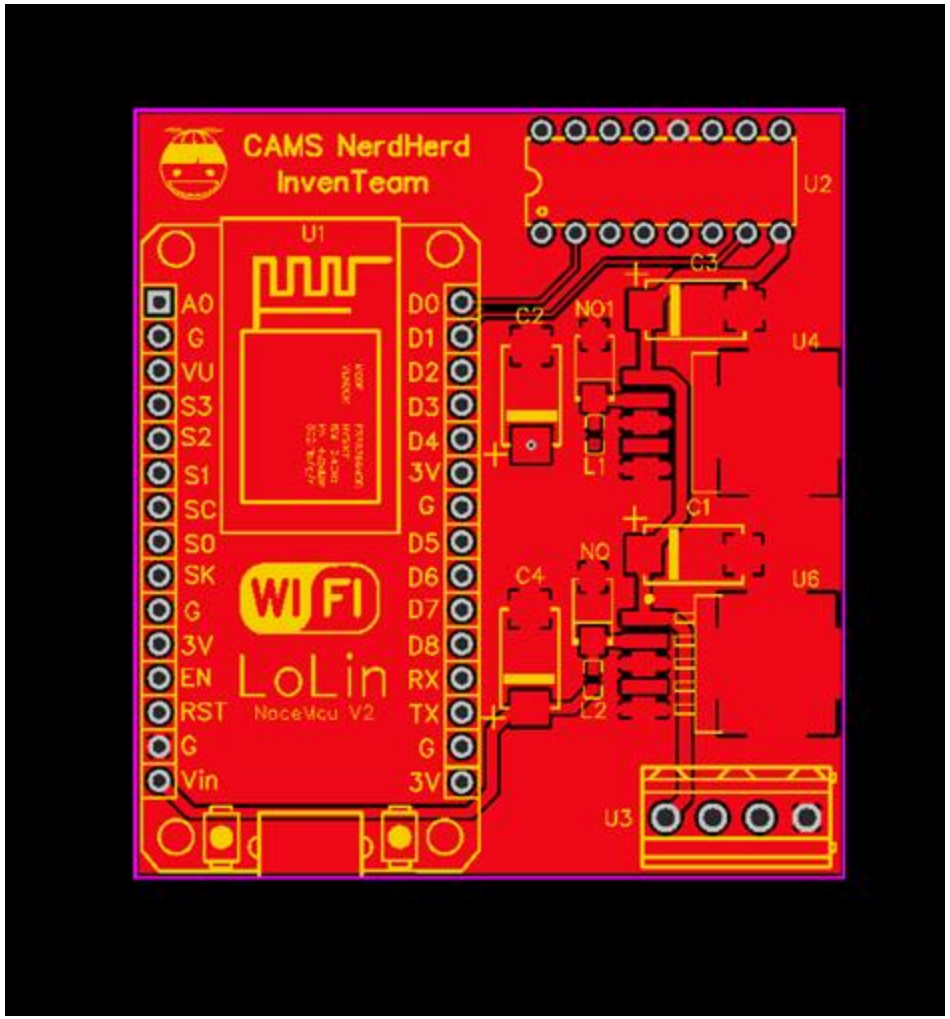
Sustainability



UPDATE WITH NEW

Moving Forward

- Electrical Improvements
- Communication Improvements
- Improved Sensor Integration
- Underwater Testing of Inflation

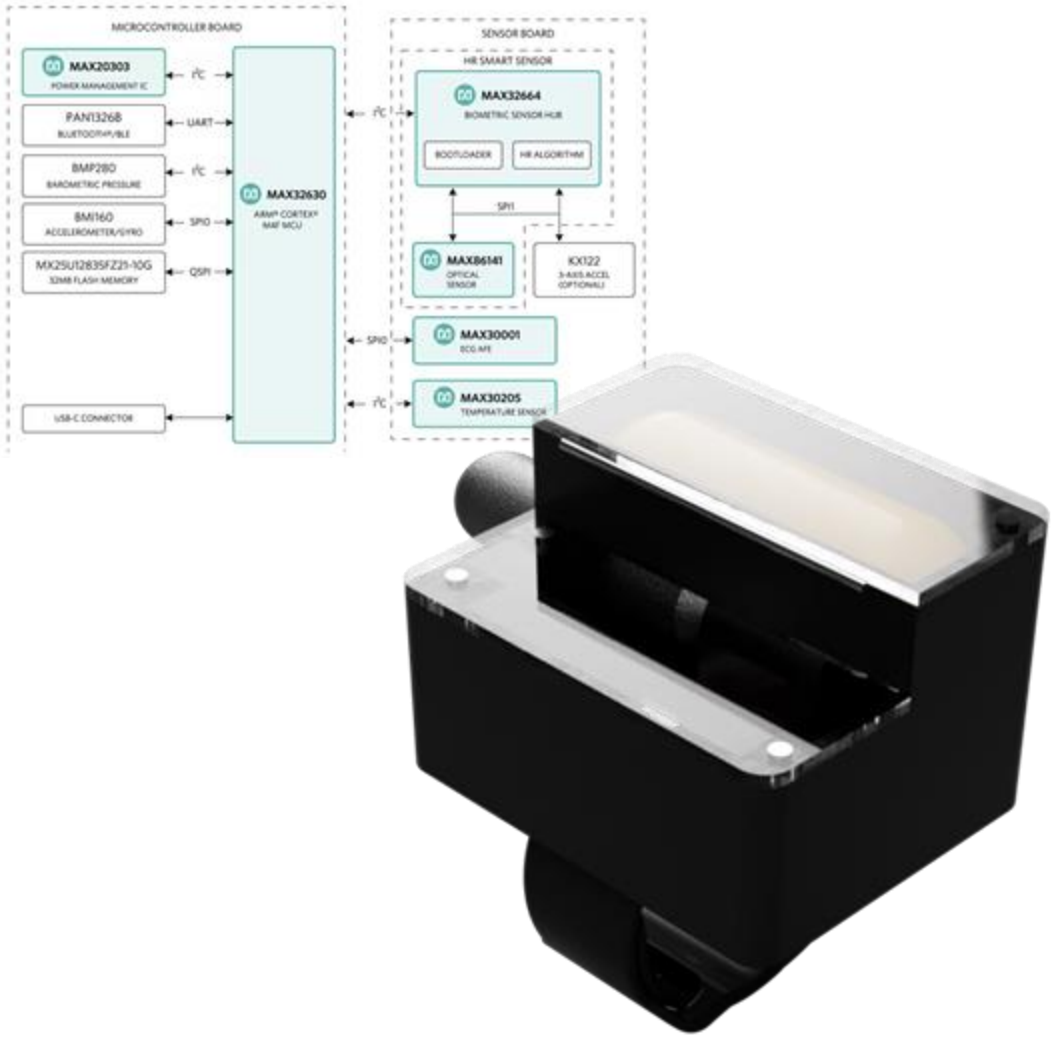


Integrated electronics PCB design

Electrical Improvements

- Smaller packaging (Decrease by 75%)
- Designed custom PCB
- Manufacturing

JLCPCB



Improved Sensor Integration

- Body temperature, pulse oximetry, and IMU data
- Previously tested algorithms on a faster microprocessor (MAX32630)

Communication Improvements

- Switch communication to bluetooth
- Easier for most users to interact

Underwater Inflation Testing

- Waterproofing electronics in *a fully integrated product* is an inherently difficult task
- Smaller footprint & watertight seal required
- Further proves design viability

Acknowledgements

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- Parents

Thank You!

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